Prompts Log: Drought in Mykolaiv Oblast

Test task: use open data to show spatially in which years drought occurred in Mykolaiv Oblast; present the result for farmers in a presentation of up to 4 slides.

AI assistant used: Claude (Anthropic), in chat. **Period:** 21-24 October 2026.
What the AI was used for: explaining the method, writing and reviewing the Python / Google Earth Engine code (Steps 1-7), checking results, planning the methodology document, and writing the slide text and recommendations.
How to read this log: prompts are reproduced word for word, in the order they were sent, including my typing mistakes. Files I attached and results I pasted are shown in square brackets, not copied. Part 1 holds the main prompts; Part 2 holds the prompts from the code-review, results-check and presentation stages.

Part 1. Project, method, code and methodology

Prompt 1

Поясни детальніше, як виконувати тестове завдання та як краще і чому буде його реалізувати:
Тестове:
За допомогою відкритих даних просторово показати, у які роки спостерігалась посуха в Миколаївській області. Результат представити у презентації не більше 4 слайдів. Уявіть, що цільова аудиторія презентації - фермери.
Результат: Презентація до 4-х слайдів; PDF з описом методології; PDF із текстом промтів (якщо використовувався AI).

Prompt 2

Clearly explain each part of the work and why we are doing it this way.

Prompt 3

Project description: Use open data to provide a spatial visualization of the years in which drought occurred in the Mykolaiv region.
Follow the plan below to create project:
Step 1: Earth Engine Setup. Authenticate, and initialize it with a project
Step 2. Define the area.
Load the border of Mykolaiv Oblast and its districts from a public boundaries dataset.
Step 3. Keep only farmland.
Use a land-cover map (ESA WorldCover) to hide cities, water and forest.
Step 4. Calculate two indicators for each year (2000-2025), April to September.

* Water balance = rain minus the amount of water that could evaporate. Negative means a dry season. This is the cause.
* Vegetation greenness (NDVI) from the MODIS satellite. Lower greenness means plants are struggling. This is the effect.

Step 5. Turn raw values into “how unusual was this year”.
For each pixel we compute a z-score: (this year’s value minus the average) divided by the typical year-to-year variation. A z-score of 0 is normal, -1 is noticeably dry/poor, -2 is extreme.
Step 6. Classify the drought.
Convert z-scores into five simple categories: normal, weak, moderate, severe, extreme.
Step 7. Count and tabulate.
For each year compute the share of farmland in drought, and the average z-score per district.
Step 8. Draw the maps.
One map per year with the same colour scale for all, a map of how many years each place had drought, and a bar chart per year.
Step 9. Validate.

Prompt 4

Could you explain—in detail, yet as simply as possible, so that even someone completely unfamiliar with the subject could understand—what we've already done in the code, and what still remains to be done?

Prompt 5

What needs to be checked manually to avoid errors?

Prompt 6

How you count every coefficient only for one year? Tell it in really easy words.

Prompt 7

Is summing water balance a good decision, and why?

Prompt 8

How do we calculate the water balance score, and what does that mean? Why do we use potential evaporation, not something else?

Prompt 9

How do we calculate all z-scores and what they mean?

Prompt 10

How do we calculate major drought years?

Prompt 11

How do we know the percentage of land in every class?

Prompt 12

Recreate step 6 to save all files ONLY localy and think about this problem: Failed to download step6_drought_2007.png: HTTP Error 400: Bad Request before doing this task

Prompt 13

Check the whole code one more time for any problems

Prompt 14

I also need code for step 7 to check the correctness of the data for misses, invalid data, and other tests that you would consider as crucial.

Prompt 15

Find what kind of limitations (data resolution, satellite estimation errors, NDVI also depends on the crop, not just on drought) accures in our project

Prompt 16

I need to make a methodology PDF. Give me the detailed planned structure of it

Prompt 17

Here is my file with the methodology. Please evaluate it critically and suggest improvements if necessary.

[attached: the methodology file]

Part 2. Code review, results check and presentation

Prompt 18

For now, these are all my files and the results they produced. Please check the accuracy of the output data; if necessary, correct the scripts and send me the complete, updated versions.

[attached: code and outputs of Steps 1-6]

Prompt 19

I need to make 4 slides for the presentation. Make a plan for slides. It should correspond to our project. data. But you need to take into consideration that this presentation will be used to show it to the farmers. They don't know data science jargon and need the easiest descriptions

Prompt 20

Make a text for Gamma to automatically create slides

Prompt 21

I don't like some of the recommendations from the final slide.
Analyze this article and choose the best practice from there and write new short recommendations based on that:
https://a7d.com.ua/agropoltika/agri-work/5399-posuhi-yak-yim-zapodyati.html

Prompt 22

Find more fresh articles with recommendations on how to deal with drought. Send me links and write new recommendations for slide 4

Prompt 23

Make a PDF with prompt text. This is what logged my assistant. Change it if needed

[attached: the PROMPTS.md log kept by the assistant]